

HDI Predictor Project - Design Document

Prepared for the SmartBridge internship submission.

System Architecture Flow: User enters input values in the web form, the backend validates them, scales the features, passes them to the trained regression model, and returns a predicted HDI score and tier.

Data Flow Logic: Missing values are imputed, features are standardized, model training is performed with train-test split, and predictions are rounded before display.

User Interface Mockups: Home page, prediction form, result page, and insights page are implemented using Flask templates and CSS styling.